

The large sample size fallacy

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Background: Significance in the statistical sense has little to do with significance in the common practical sense. Statistical significance is a necessary but not a sufficient condition for practical significance. Hence, results that are extremely statistically significant may be highly nonsignificant in practice. The degree of practical significance is generally determined by the size of the observed effect, not the p-value. The results of studies based on large samples are often characterized by extreme statistical significance despite small or even trivial effect sizes. Interpreting such results as significant in practice without further analysis is referred to as the large sample size fallacy in this article.

Aim: The aim of this article is to explore the relevance of the large sample size fallacy in contemporary nursing research. **Results:** Relatively few nursing articles display explicit measures of observed effect sizes or include a qualitative discussion of observed effect sizes. Statistical significance is often treated as an end in itself.

Conclusion: Effect sizes should generally be calculated and presented along with p-values for statistically significant results, and observed effect sizes should be discussed qualitatively through direct and explicit comparisons with the effects in related literature.

Keywords: effect size, nursing research, practical significance, p-value, statistical significance.

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Introduction

A common problem in nursing research is that statistical significance is treated as almost synonymous with practical significance in discussions, conclusions and recommendations. However, statistical significance of the results – differences or relations – does not necessarily mean that they have practical importance or consequence. Statistical significance can be viewed as a necessary but not as a sufficient condition for practical significance. The degree of practical significance is instead determined by the effect size, not the p-value. In a nutshell, an effect size describes the observed relative divergence from the expectation specified in the null hypothesis. For example, when the null hypothesis specifies that two population means are equal, the effect size is usually measured as the observed difference between the two means divided by the standard deviation of the data.¹

As effect sizes (unlike p-values) are not directly affected by sample sizes, a statistically significant result can be obtained despite a small effect size as long as the sample is

large enough. Similarly, large effect sizes do not guarantee statistical significance when small samples are used. The superiority of large samples over small samples, *ceteris paribus*, is probably one of the least controversial truths in research, and the aim of this article is certainly not to claim otherwise. However, as large enough sample sizes always lead to rejection of the null hypothesis (2), the results of studies based on large samples are often characterized by extreme statistical significance despite small or even trivial effect sizes. In such cases, statistical significance does not necessarily imply practical significance. The present article discusses this problem.

The article is organized as follows. First, we extend the discussion about the relation between effect sizes and statistical significance. The next section discusses the magnitude of the problem based on quantitative and qualitative findings from studies published in two prestigious nursing journals. The final section concludes the article.

Effect sizes and statistical significance

A p-value is a measure of the probability of obtaining a test statistic at least as extreme as the observed one, conditional on a true null hypothesis. Ideally, we would prefer a measure of the probability of a true null hypothesis conditional on the observed data; however, such a measure cannot be created (3). Instead, the p-value

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¹ This specific effect size measure is called Cohen's *d* (1).

Table 1 Effect size indexes and benchmark values (based on 5)

Test	Effect size index	Benchmark		
		Small	Medium	Large
t-test for independent means	$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}$	0.20	0.50	0.80
z-test for independent proportions	$h = 2 \arcsin \sqrt{p_1} - 2 \arcsin \sqrt{p_2}$	0.20	0.50	0.80
Correlation	r	0.10	0.30	0.50
Multiple correlation	$f^2 = \frac{R^2}{1-R^2}$	0.02	0.15	0.35
One-way ANOVA ^a	$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{Total}}}$	0.01	0.09	0.25
Chi-square test for independence (k is the smallest of the number of rows and the number of columns in the cross tab)	χ^2	0.10	0.30	0.50
	$v = \sqrt{\frac{\chi^2}{n(k-1)}}$	0.07	0.21	0.35
		0.06	0.17	0.29

^aLevine & Hullett (6) notes that η^2 has many desirable properties and will therefore be used here instead of f suggested by Cohen (5). Since the square root of η^2 is equal to r when there is one degree of freedom in the numerator, the benchmark values for η^2 are equal to the squared benchmark values for r . Cohen's f has the advantage that it can be calculated based on individual sample sizes, means and standard deviations.

is generally used, although often incorrectly. Common misconceptions about what the p-value measures include (i) the probability that the results occurred because of chance, (ii) the probability that the results are possible to replicate, (iii) the probability that the null hypothesis is false, (iv) the probability that the results are important and (v) the probability that the results can be generalized (3). It is true that a small p-value is an indication of an actual difference or relation; however, it provides little indication about the magnitude of the difference or the strength of the relation. We need effect size measures for determining that.

Effect size measures are easy to calculate. They can, however, be difficult to interpret. In general, we should interpret our effects in direct and explicit comparison with the effects in related prior literature (4). Moreover, there are general benchmarks that can be employed when such comparisons cannot be made (5). In Table 1, effect size formulas and their benchmark values for some common statistical tests are summarized.

As noted earlier, the main difference between the p-value and the effect size, which is also the basis for this article, is that the effect size is not related to the sample size, while the p-value is strictly dependent on the sample size. The sample size is obviously an important factor in a quantitative study, but its relevance is too often misunderstood. For example, results based on small samples are sometimes criticized by reviewers of research reports and scientific articles because they are unsure whether statistical significance can be claimed effectively (7). However, if results are statistically significant, they are statistically significant despite the sample size. Small samples do not make statistically significant results less statistically significant. The p-value is the only sample statistic that indicates whether an observed difference or dependency is statistically significant, and the p-value

does take sample size into account.² A small sample does, however, provide less confidence on the true effect size. While effect sizes are easy to point estimate, as shown in Table 1, exact formulas for confidence limits of the true effect size based on sample data are generally not available because of the mathematical circular reasoning that would be required. However, computer-based iteration procedures can be invoked to obtain reasonable guesstimates of such confidence limits (4). Nevertheless, a larger sample size is fundamentally linked to narrower confidence intervals of the true population parameters including the effect size. Hence, a larger sample size is better than a smaller one, although only if the results are interpreted correctly.

The hidden advantage of working with small sample sizes is that statistical significance is fundamentally related to practical significance in terms of larger effect sizes. In other words, statistically significant results will generally not be reached in a small sample study unless the effect sizes are large. Assume, for example, that a clinical trial is conducted, where 20 subjects receive both treatment X and treatment Y and are required to convey their preference between them. Assume that 15 subjects preferred treatment X. Now, a regular chi-squared test on these data would indicate a statistically significant difference between X and Y ($\chi^2 = 5.00$, $p = 0.0253$), with a very large observed effect size of $h = 1.05$. If only 14 subjects had preferred treatment X, the difference would not have been statistically significant ($\chi^2 = 3.20$, $p = 0.0736$), despite the large observed effect size of $h = 0.82$. Hence, the practical significance of statistically significant results from small sample studies is easier to interpret; that is, statistical

² Whether p-values are the logical basis for statistical inference at all is another interesting issue, albeit not discussed here. See, for example, (8).

Table 2 Results from six hypothetical clinical trials

Trial	n	Preference for X	% with preference for X	p-value	h	95% CI	
						Lower limit	Upper limit
1	20	15	75.00	0.0253	1.05	56.02	93.98
2	80	50	62.50	0.0253	0.51	51.89	73.11
3	180	105	58.33	0.0253	0.33	51.13	65.54
4	1125	600	53.33	0.0253	0.13	50.42	56.25
5	21 125	10 725	50.77	0.0253	0.03	50.10	51.44
6	1 997 120	1 000 140	50.08	0.0253	<0.01	50.01	50.15

significance will generally not be found unless the effect size is large enough to be significant in practice.

Again, this does not imply that there is something wrong with large sample sizes. As noted previously, large samples are always better than small samples, *ceteris paribus*, because all parameter values (including effect sizes) can be estimated with higher accuracy. Some statistical methods also require at least a certain sample size because otherwise the test statistic cannot be assumed to follow a certain distribution. Moreover, a sample of at least a certain size is generally needed to achieve sufficient power. In many studies, for example, drug studies or cardiovascular device studies, a large sample is needed because even small effect sizes are often useful for detection. In such cases, sample size calculations actually serve the purpose of reaching a certain power level. However, an important objection to this is that it must be ensured that a statistically significant result from a study based on a large sample possesses a large enough effect size to be practically significant. Statistical significance, in itself, may have little meaning. Assume, for example, that the hypothetical clinical trial above was conducted with 21 125 subjects and that 10 725 (50.77%) expressed preference for treatment X. A regular chi-squared test on these data would indicate a statistically significant difference between X and Y ($\chi^2 = 5.00$, $p = 0.0253$). However, despite the statistical significance, the difference is probably trivial in practice because the effect size is very small, that is, $h = 0.03$. Hence, without some additional evidence that such a small effect has practical relevance, it should probably not be highlighted as an important result of the study, despite the statistical significance.

The hypothetical examples above illustrate the relation between sample size, p-value, and effect size. In table 2, data from six different examples of such clinical trials are summarized. Based on chi-square tests, they all have the same p-value, but a small sample size obviously corresponds with a large effect size, while a large sample size corresponds with a small effect size. Thus, a p-value that indicates statistical significance may mean anything from a very large effect size to an infinitesimally small effect size. It depends on the actual sample size. These data also explain why results tend to be either extremely statistically

significant or not significant at all when large samples are used. When the sample size increases, the effect size interval where the p-value can fall, for example, between 0.05 and 0.01, reduces.

Again, it should be stressed that this is not an argument in favour of small sample sizes. As noted earlier, large samples are always better than small samples, *ceteris paribus*. Table 2 clearly shows how confidence intervals become narrower when larger samples are used, providing increased precision of effect estimates. Rather, it is an illustration of the fact that the p-value alone is not a useful statistical measure given a large sample size (9). When the sample size is large enough, all differences and dependencies will be statistically significant, even though the effect sizes may be too small to have any practical significance. Disregarding this problem is what we refer to as the large sample size fallacy in this article. Hence, research should report effect sizes or similar measures (e.g. confidence intervals) and provide their qualitative interpretations, especially when large samples are used.

The magnitude of the problem in nursing research

To explore the magnitude of this problem in contemporary nursing research, recent empirical research published in two prestigious nursing journals (the Scandinavian Journal of Caring Sciences and the Journal of Advanced Nursing) was reviewed. In the Scandinavian Journal of Caring Sciences, during 2011, statistical results based on p-values were presented in 25 of 86 empirical studies. In the Journal of Advanced Nursing, during 2010, 92 of 188 research papers relied on statistical results based on p-values.³ These 117 statistics-based articles were reviewed in detail with respect to sample sizes and the treatment of effect sizes.

³ A few articles presented p-values primarily to describe degrees of model fit in multivariate analyses. There were also a number of articles based on descriptions, but no statistical analyses based on p-values (even though such analyses seemed quite plausible in several cases) of the quantitative data. None were included.

After manual calculation of actual effect sizes in all cases where they were not explicitly displayed by the authors, the articles were categorized based on the following criteria:

1. Whether various measures of observed effect sizes or measures with similar informational content, for example, confidence intervals for differences, were explicitly displayed for all statistically significant results. (Hence, e.g. odds ratios or values of r or R^2 were not considered as measures of effect sizes in this context if the study presented other types of statistically significant results where no measure of effect size is displayed.)
2. Whether observed effect sizes in some way were discussed qualitatively, regardless of whether their measures were explicitly displayed or not, in relation to the fact that effects were found to be statistically significant. (Note: Mentioning a small effect as a limitation of the study was not considered as a qualitative discussion.)
3. Whether the effect size was specified in sample size calculations.
4. The lowest actual effect size related to an explicitly presented statistically significant result.

Note that these categories are used simply to describe how the effect size issue is treated in contemporary nursing research. Hence, they do not suggest that every nursing research article should include measures and discussions of observed effect sizes. Nor do they suggest that small effect sizes are of little or no relevance in practice, or that sample sizes should always be calculated based on effect sizes. In some study types, for example, equivalence studies, the presence of a statistically significant difference irrespective of its magnitude is the only relevant issue to be addressed, and in some other cases, small effect sizes may be highly significant in practice. Generally, however, it is considered good practice to clarify the estimated size of an observed empirical effect and the practical significance of an effect of that size.

The median sample size of the 117 reviewed studies was 173. As several studies used very large samples, the mean sample size was as high as 504. Table 3 presents the distribution of different sample sizes.

As table 3 shows, very large samples (defined here as $n > 500$) were quite common. Thus, there was an obvious potential for the large sample size fallacy. Table 4 shows the number of articles displaying explicit measures of

Table 3 Sample sizes, frequencies

Sample size	Scandinavian Journal of Caring Sciences	Journal of Advanced Nursing
<30	0	3
31–100	9	19
101–500	7	47
>501	9	23

Table 4 Treatment of effect sizes, frequencies

Scandinavian Journal of Caring Sciences		Journal of Advanced Nursing	
Displays ES measures	Includes qualitative discussion of observed ES	Displays ES measures	Includes qualitative discussion of observed ES
No	21	20	68
Yes	4	5	24

observed effect sizes (or measures with similar informational content) and the number of articles including qualitative discussions on observed effect sizes.

A majority of the articles did not display explicit measures of observed effect sizes. In addition, only 14 of the 117 articles included some qualitative discussion of observed effect sizes. That is, in majority of the cases, statistical significance was considered an end in itself.

In table 5, the effect sizes specified or implied in sample size calculations and the lowest actual effect sizes for explicitly presented statistically significant result are displayed. In 87 of the 117 cases, effect size was not used to justify the sample sizes. It should be noted that in quite a few of these cases, the authors correctly assumed or implied that the specific sampling strategy made the effect size perspective irrelevant. A typical example is (10) where the authors chose to invite all individuals in the population of interest to participate because their number was small. In other cases, however, the authors incorrectly assumed that their sample sizes were so large that effect sizes did not need to be considered. In yet other cases, the effect size perspective was simply ignored in the section where sample size was discussed.

A majority of those who did use effect size to justify their sample sizes chose the medium size. However, the lowest actual effect size for an explicitly presented statistically significant result was small in 45 of the 117 cases. In nine of these 45 cases, the authors uncritically presented at least one statistically significant result with a small effect size even though they explicitly based their study on a medium effect size in the sample size calculation. Furthermore, of the 32 studies with sample sizes above 500, 24 presented statistically significant results with small effect sizes while omitting both measures and qualitative discussions of observed effect size. In a few cases, effect sizes could not be calculated because of insufficient information.⁴

⁴ Given the theme of this article, it must be clarified that effect sizes are not used to describe the results in this study because the data represent case studies rather than random samples. As we cannot use these data to test whether there is a statistically significant effect, it is not relevant to measure the size of it.

Table 5 Used and found effect sizes, frequencies

ES	Scandinavian Journal of Caring Sciences		Journal of Advanced Nursing	
	In sample size calculation	Lowest actual ES for a statistically significant result	In sample size calculation	Lowest actual ES for a statistically significant result
n/a	24	5	63	7
Small	0	11	1	34
Medium	1	9	21	35
Large	0	0	7	16

Although some authors (e.g. 11) showed explicit awareness of the large sample size fallacy, several articles in both journals were found to uncritically present statistically significant results characterized by small or very small effect sizes after a large sample study had been conducted. For example, Liou et al. (12) used a very large sample size ($n = 8,640$) in a study that examined risk factors for adolescent obesity. The authors emphasize in their conclusions that watching television for more than 2 hours per day and sleeping for <7.75 hours per day are factors that increase the risk of obesity among adolescents. These factors were indeed statistically significant owing to the large sample size, but they were also characterized by effect sizes that were clearly below the benchmark value for a small effect (effects were found to range from 0.036 to 0.085 based on the data in the article). Paternal as well as maternal obesity was also found to be associated with adolescent obesity, in both cases with p -values below 0.001, but the effect sizes were again found to be very small and clearly below the benchmark value for a small effect in both cases. Furthermore, the authors also emphasized a statistically significant reciprocal relationship of cause and effect between having nightly snacks and obesity, although the effect size was clearly below the benchmark value for a small effect. Finally, at least 10 reported significant differences between male and female subjects, or obese and nonobese subjects, were characterized by no observed difference in median value, indicating that outliers were responsible for the statistically significant difference.

Another example is Xu et al. (13) where a very large number of subjects ($n = 5226$) were used to understand self-rated health and its associated factors in Chinese adolescents. The authors found a significant negative bivariate correlation between age and health level (erroneously reported with a positive value of Pearson's r in the text). Even though the p -value was below 0.05 owing to the large sample size, the correlation was extremely weak ($r = -0.027$). Furthermore, in the subsequent multivariate

analysis, age was positively correlated with health level, but the inconsistency was ignored by the authors. In addition, the degree of regression model fit (i.e. the coefficient of determination) should have been reported as the effect size otherwise was entirely hidden. Finally, although it was not related to the large sample size fallacy, it should be mentioned that the variable 'family type' clearly was nominal with three possible outcomes and should not have been transformed to a quantitative scale, as done by the authors in order to include it in the regression model.

Several studies characterized by the hidden advantage of small samples were also found. For example, Szeto et al. (14) reported the results of a pilot study conducted to investigate the effect of a tailor-made ergonomic intervention programme for community nurses. The study was based on an intervention group ($n = 14$) and a control group ($n = 12$). Of course, the power of a study with such a small sample size was low, but the results were interesting in terms of effect size (although such measures were not explicitly displayed in the article) – every statistically significant result was characterized by a large effect size.

Finally, it should be noted that numerous typographical or logical errors related to the general handling of statistical data were found during the reviews. For example, p -values equal to or even below zero were reported; statistical significance was defined as $p > 0.05$ in one case; in some cases, confidence intervals for individual variables were reported instead of their difference, although their difference was researched; the magnitudes or signs of reported test statistics were clearly incorrect in several cases; statistically nonsignificant results were addressed in a population perspective and the choices of statistical tests were inaccurate in several cases. Hence, both authors and reviewers should be more thorough regarding the presentation of statistical results in nursing research.

Conclusion

Contemporary nursing research with a statistical approach seems to rely heavily on p -values. However, statistical significance is rarely an end in itself. Without some additional analyses of the effect that created the statistical significance at the current sample size, the practical value of a statistically significant difference or a dependency is in most cases impossible to assess. As statistical significance is fundamentally related to sample size while effect size does not depend on sample size, a strict focus on the binary question of whether statistical significance has been found is misleading. When the sample size is large enough, all differences and dependencies will be statistically significant, even though the effect sizes may be too small to have any practical significance. Disregarding this problem is what we refer to as 'the large sample size fallacy' in this

article. As we have seen, researchers who use statistical methods to analyse data in nursing research often seem to omit measures of effect sizes as well as qualitative discussions around them, thereby exposing themselves to this fallacy.

The remedy is actually rather simple. When a result is found to be statistically significant, the effect size along with the p-value must be reported, especially in studies based on large samples, and a brief interpretation of its practical relevance in relation to prior literature must be provided.

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